

Kai Zhang

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I am a third-year CS Ph.D. student advised by Prof. Yu Su. I have published papers in top NLP/CV/ML venues, many of which have been nominated for or won Best Paper Awards (3), or have been featured as Oral (8) and Spotlight (2) presentations. My work has 3,000 citations and an h-index of 19.

EDUCATION

The Ohio State University Ph.D. in Computer Science and Engineering • GPA: 3.90/4.0 Advisor: Yu Su	Ohio, USA Aug 2022-Jun 2027 (Expected)
Zhejiang University City College B.E. in Computer Science and Technology • GPA: 3.99/4.0 Rank: 1/118	Zhejiang, China Sept 2016-Jun 2020

RESEARCH EXPERIENCE

Meta GenAI Research Scientist Intern; Llama Multimodal and Agents Teams	New York, USA May 2025-Present
Microsoft Research Research Intern; Host: Chenglong Wang, Jeevana Priya Inala, and Jianwei Yang	Washington, USA May 2024-Aug 2024
Google DeepMind Student Researcher; Host: Yi Luan, Hexiang (Frank) Hu, and Ming-Wei Chang	Washington, USA Aug 2023-Mar 2024
Microsoft Research Research Intern; Host: Chongyang Tao and Tao Shen	Beijing, China Mar 2022-July 2022
Minds, Machines, and Society Group, Dartmouth College Research Intern; Advisor: Soroush Vosoughi	New Hampshire, USA Jun 2020-Sept 2020
THU NLP Group, Tsinghua University Research Assistant; Advisor: Zhiyuan Liu	Beijing, China Nov 2019-Mar 2021

SELECTED PUBLICATIONS

(† indicates my mentee. See Google Scholar)

- MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions**
Kai Zhang, Yi Luan, Hexiang Hu, Kenton Lee, Siyuan Qiao, Wenhui Chen, Yu Su, and Ming-Wei Chang
In ICML (**Oral**), 2024.
Used as Training Data for Gemini Embedding; Widely Used in Snap, Pinterest, and Google
- TravelPlanner: A Benchmark for Real-World Planning with Language Agents**
Jian Xie†*, Kai Zhang*, Jiangjie Chen, Tinghui Zhu, Renze Lou, Yuandong Tian, Yanghua Xiao, and Yu Su
In ICML (**Spotlight**), 2024. (*: Equal Contribution)
- MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI**
Xiang Yue*, Yuansheng Ni*, Kai Zhang*, Tianyu Zheng*, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, and Wenhui Chen
In CVPR (**Oral; Best Paper Finalist**), 2024. (*: Core Contribution)
2M Total Downloads; Used by Google DeepMind, OpenAI, Anthropic, xAI, NVIDIA, Meta, and Microsoft

4. ***Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts***
Jian Xie†*, Kai Zhang*, Jiangjie Chen, Renze Lou, and Yu Su
In ICLR (**Spotlight**), 2024. (*: Equal Contribution)
5. **MUFFIN: Curating Multi-Faceted Instructions for Improving Instruction-Following**
Renze Lou†, Kai Zhang, Jian Xie, Yuxuan Sun, Janice Ahn, Hanzi Xu, Yu Su, and Wenpeng Yin
In ICLR, 2024.
6. **MagicBrush: A Manually Annotated Dataset for Instruction-Guided Image Editing**
Kai Zhang*, Lingbo Mo*, Wenhui Chen, Huan Sun, and Yu Su
In NeurIPS Datasets and Benchmarks, 2023. (*: Equal Contribution)
7. **Aligning Instruction Tasks Unlocks Large Language Models as Zero-Shot Relation Extractors**
Kai Zhang, Bernal Jiménez Gutiérrez, and Yu Su
In Findings of ACL, 2023.
8. **LED: Lexicon-Enlightened Dense Retriever for Large-Scale Retrieval**
Kai Zhang, Chongyang Tao, Tao Shen, Can Xu, Xiubo Geng, Binxing Jiao, and Daxin Jiang
In WWW (**Oral**), 2023.
9. **Contributions of Transformer Attention Heads in Multi- and Cross-lingual Tasks**
Weicheng Ma*, Kai Zhang*, Renze Lou, Lili Wang, and Soroush Vosoughi
In ACL (**Oral**), 2021. (*: Equal Contribution)
10. **Open Hierarchical Relation Extraction**
Kai Zhang*, Yuan Yao*, Ruobing Xie, Xu Han, Zhiyuan Liu, Fen Lin, Leyu Lin, and Maosong Sun
In NAACL-HLT, 2021. (*: Equal Contribution)

OTHER PUBLICATIONS

1. **MMMU-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark**
Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhui Chen, and Graham Neubig
In ACL, 2025.
2. **CPath-Omni: A Unified Multimodal Foundation Model for Patch and Whole Slide Image Analysis in Computational Pathology**
Yuxuan Sun, Yixuan Si, Chenglu Zhu, Xuan Gong, Kai Zhang, Pingyi Chen, Ye Zhang, Zhongyi Shui, Tao Lin, and Lin Yang
In CVPR, 2025.
3. **AAAR-1.0: Assessing AI’s Potential to Assist Research**
Renze Lou, Hanzi Xu, Sijia Wang, Jiangshu Du, Ryo Kamoi, Xiaoxin Lu, Jian Xie, Yuxuan Sun, Yusen Zhang, Jihyun Janice Ahn, Hongchao Fang, Zhuoyang Zou, Wenchao Ma, Xi Li, Kai Zhang, Congying Xia, Lifu Huang, and Wenpeng Yin
In ICML, 2025; AI4Research@AAAI (**Best Paper Award**), 2025.
4. **PathGen-1.6M: 1.6 Million Pathology Image-Text Pairs Generation through Multi-Agent Collaboration**
Yuxuan Sun, Yunlong Zhang, Yixuan Si, Chenglu Zhu, Zhongyi Shui, Kai Zhang, Jingxiong Li, Xingheng Lyu, Tao Lin, and Lin Yang
In ICLR (**Oral**), 2025.
5. **MuirBench: A Comprehensive Benchmark for Robust Multi-image Understanding**
Fei Wang, Xingyu Fu, James Y Huang, Zekun Li, Qin Liu, Xiaogeng Liu, Mingyu Derek Ma, Nan Xu, Wenxuan Zhou, Kai Zhang, Tianyi Lorena Yan, Wenjie Jacky Mo, Hsiang-Hui Liu, Pan Lu, Chunyuan Li, Chaowei Xiao, Kai-Wei Chang, Dan Roth, Sheng Zhang, Hoifung Poon, and Muhao Chen
In ICLR, 2025.
6. **Revealing the Barriers of Language Agents in Planning**
Jian Xie, Kexun Zhang, Jiangjie Chen, Siyu Yuan, Kai Zhang, Yikai Zhang, Lei Li, and Yanghua Xiao
In NAACL, 2025.

7. **PathMMU: A Massive Multimodal Expert-Level Benchmark for Understanding and Reasoning in Pathology**
Yuxuan Sun, Hao Wu, Chenglu Zhu, Sunyi Zheng, Qizi Chen, Kai Zhang, Yunlong Zhang, Xiaoxiao Lan, Mengyue Zheng, Jingxiong Li, Xinheng Lyu, Tao Lin, and Lin Yang
In ECCV (**Oral; Best Paper Finalist**), 2024.
8. **Deductive Beam Search: Decoding Deducible Rationale for Chain-of-Thought Reasoning**
Tinghui Zhu†*, Kai Zhang*, Jian Xie, and Yu Su
In COLM, 2024. (*: Equal Contribution)
9. **How Easily do Irrelevant Inputs Skew the Responses of Large Language Models?**
Siye Wu, Jian Xie, Jiangjie Chen, Tinghui Zhu, Kai Zhang, and Yanghua Xiao
In COLM, 2024.
10. **ImagenHub: Standardizing the Evaluation of Conditional Image Generation Models**
Max Ku, Tianle Li, Kai Zhang, Yujie Lu, Xingyu Fu, Wenwen Zhuang, and Wenhui Chen
In ICLR, 2024.
11. **Large Language Model Instruction Following: A Survey of Progresses and Challenges**
Renze Lou†, Kai Zhang, and Wenpeng Yin
In Computational Linguistics, 2024.
12. **CIF-Bench: A Chinese Instruction-Following Benchmark for Evaluating the Generalizability of Large Language Models**
Yizhi Li, Ge Zhang, Xingwei Qu, Jiali Li, Zhaoqun Li, Zekun Wang, Haoyuan Li, Ruibin Yuan, Yi Ma, Kai Zhang, Wangchunshu Zhou, Yiming Liang, Lei Zhang, Lei Ma, Jiajun Zhang, Zuowen Li, Stephen W. Huang, Chenghua Lin, Wenhui Chen, and Jie Fu
In Findings of ACL, 2024.
13. **UMIE: Unified Multimodal Information Extraction with Instruction Tuning**
Lin Sun*, Kai Zhang*, Qingyuan Li†*, and Renze Lou
In AAAI (**Oral**), 2024. (*: Equal Contribution)
14. **PathAsst: Redefining Pathology through Generative Foundation AI Assistant for Pathology**
Yuxuan Sun, Chenglu Zhu, Sunyi Zheng, Kai Zhang, Zhongyi Shui, Xiaoxuan Yu, Yizhi Zhao, Honglin Li, Yunlong Zhang, Ruojia Zhao, Xinheng Lyu, and Lin Yang
In AAAI (**Oral**), 2024.
15. **Automatic Evaluation of Attribution by Large Language Models**
Xiang Yue, Boshi Wang, Ziru Chen, Kai Zhang, Yu Su, and Huan Sun
In Findings of EMNLP, 2023.
16. **UnifieR: A Unified Retriever for Large-Scale Retrieval**
Tao Shen, Xiubo Geng, Chongyang Tao, Can Xu, Guodong Long, Kai Zhang, and Daxin Jiang
In KDD, 2023.
17. **ChatDoctor: A Medical Chat Model Fine-Tuned on a Large Language Model Meta-AI (LLaMA) Using Medical Domain Knowledge**
Yunxiang Li, Zihan Li, Kai Zhang, Ruilong Dan, and You Zhang
In Cureus, 2023.
18. **GradTS: A Gradient-Based Automatic Auxiliary Task Selection Method Based on Transformer Network**
Weicheng Ma, Renze Lou, Kai Zhang, Lili Wang, and Soroush Vosoughi
In EMNLP, 2021.
19. **Unified and Incremental SimRank: Index-free Approximation with Scheduled Principle**
Fanwei Zhu, Yuan Fang, Kai Zhang*, Kevin Chen-Chuan Chang, Hongtai Cao, Zhen Jiang, and Minghui Wu
In IEEE TKDE, 2021. (*: First Student Author)
20. **Adversarial Language Games for Advanced Natural Language Intelligence**
Yuan Yao, Haoxi Zhong, Zhengyan Zhang, Xu Han, Xiaozhi Wang, Kai Zhang, Chaojun Xiao, Guoyang Zeng, Zhiyuan Liu, and Maosong Sun
In AAAI, 2021.

21. RpBERT: A Text-image Relation Propagation-based BERT Model for Multimodal NER

Lin Sun*, Jiquan Wang†*, Kai Zhang, Fangsheng Weng, and Yindu Su
In AAAI, 2021. (*: Equal Contribution)

22. TOI-CNN: A Solution of Information Extraction on Chinese Insurance Policy

Lin Sun*, Kai Zhang*, Fule Ji, and Zhenhua Yang
In NAACL-HLT, 2019. (*: Equal Contribution with Advisor)

SELECTED HONORS & AWARDS

Best Paper Award, AI4Research Workshop@AAAI	2025
Best Paper Finalist, ECCV	2024
Best Paper Finalist, CVPR	2024
Valedictorian (Top 1), Zhejiang University City College	2020
National Scholarship (Top 1%), Ministry of Education of China	2019

INVITED TALKS

MagicLens: Next-Generation Image Retrieval Models

AI TIME Seminar
NLP Group, Zhejiang University
Beijing Academy of Artificial Intelligence

Language Agents: Grounding and Planning

AISOC Lab, Carnegie Mellon University

TravelPlanner: A Benchmark for Real-World Planning with Language Agents

Real World Journeys Group, Google

Aligning LLM Instruction Tasks for Zero-Shot Relation Extraction

NLP Seminar, University of Notre Dame
AI TIME Seminar

MENTORING

Renze Lou (ZUCC B.S. → PSU Ph.D.; 2019–2022): Instruction Tuning
Jian Xie (ZUCC B.S. → FDU M.S. → OSU Ph.D.; 2019–2024): Language Agents
Yuxuan Sun (ZUCC B.S. → ZJU Ph.D.; 2019–2024): Multimodality, Medical AI
Tinghui Zhu (FDU M.S. → UCD Ph.D.; 2023–2025): Reasoning, Multimodality

SERVICES

PC Member/Reviewer:

NLP: ARR (since 2022); CCKS; NLPCC; EMNLP; ACL; NAACL

CV: CVPR; ICCV; TPAMI

ML: NeurIPS; ICLR; ICML; COLM

Other: WWW; KDD; CIKM; SDM; UIST; AAAI; TMM

(Co-)Organizer:

T2FM Workshop: Trustworthy Study Transfers from Classical to Foundation Models@ICCV 2025